

SUBMITTED WORK

Design &
Implementation
of a Multi-Campus
University Network

Infrastructure

Apex University · Eight-Campus Network · Cisco Packet Tracer · OSPF Dynamic Routing

STUDENT SUBMISSION DETAILS

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Executive Summary

This report presents the complete design and implementation of a campus-wide network infrastructure for Apex University — a multi-campus institution modelled on a real-world enterprise environment. The network spans eight geographically distributed campuses, each equipped with wired and wireless end-user devices, interconnected through serial WAN links running OSPF dynamic routing. A centralised server farm at Campus 8 provides DHCP, DNS, and HTTP services to all campuses. The design follows industry best practices in IP subnetting, network segmentation, and hierarchical routing. Every mandatory feature has been fully implemented and verified in Cisco Packet Tracer, and this report documents the design rationale, configuration, and proof-of-functionality through annotated screenshots.

CHAPTER 1

Introduction

Modern universities are technology-driven institutions where network infrastructure is as critical as any physical facility. Lecture portals, digital libraries, administrative systems, and campus-wide Wi-Fi all depend on a reliable, scalable, and well-structured network. This project simulates precisely that environment — designing and implementing a fully functional enterprise-grade network for Apex University, a fictional institution modelled after real-world private universities in Bangladesh.

Apex University operates across eight separate campuses, each hosting classrooms, laboratories, administrative offices, and common areas. Every area requires both wired and wireless connectivity. The central campus (Campus 8) houses a dedicated server room providing DHCP, DNS, and web services to the entire institution over a serial WAN backbone. The simulation was built entirely in Cisco Packet Tracer, and every configuration in this report reflects real-world networking practices — techniques directly applicable to enterprise network engineering roles.

CHAPTER 2

Project Requirements & Compliance

The project specification defined the following mandatory requirements. The table below maps each requirement to its implementation and current status:

#	Requirement	Implementation Detail	Status
1	Minimum 8 campus LANs with wired and wireless hosts	8 campuses; each has 2960 switch, Access Point, 12+ PCs, laptops and smartphones	✓
2	All inter-router links via serial connections	WIC-2T modules added to all routers; 10 serial DCE/DTE links at 64000 bps clock rate	✓
3	Dynamic routing using OSPF	OSPF Process 1, Area 0 on all 8 routers; all networks advertised with correct wildcard masks	✓
4	Single DHCP server for all campuses	DHCP server at 10.10.80.10 with 8 address pools; ip helper-address relay on every LAN interface	✓
5	Single DNS server for university domain	DNS at 10.10.80.20; A-record: www.apex.edu.bd → 10.10.80.30; distributed via DHCP to all hosts	✓
6	University website accessible from any campus	HTTP server at 10.10.80.30; custom index.html created; tested from wireless hosts on multiple campuses	✓
7	IP addresses from all three IP classes	Class A (10.10.x.0/24), Class B (172.16.x.0/30 WAN links), Class C (192.168.x.0/24 LANs)	✓
8	Dedicated server room for all servers	DHCP, DNS and Web Server all on Campus 8 LAN segment (10.10.80.0/24)	✓
9	Follow the given physical topology exactly	All 10 serial links match the specification: R1-R8, R1-R2, R2-R8, R7-R8, R6-R8, R6-R7, R3-R4, R3-R5, R4-R5, R5-R8	✓

Table 1: Project Requirements Compliance Matrix

CHAPTER 3

Network Architecture Design

The network uses a **hierarchical three-tier design** adapted for a multi-campus university. This separates the network into distinct functional layers, making it easier to manage, secure, and scale.

3.1 Three-Tier Design Layers

Core Layer	Router 8 (Campus 8) is the backbone hub. It connects to five other routers (R1, R2, R5, R6, R7) via serial WAN links and hosts the server room. All inter-campus traffic passes through here. R8 carries five serial interfaces, requiring two extra WIC-2T modules.
Distribution Layer	Routers 1–7 serve as distribution-layer devices for their campuses. They aggregate access-layer traffic, apply DHCP relay (ip helper-address), and run OSPF to exchange routes with neighbours.
Access Layer	Cisco 2960-24TT switches on each campus connect all wired end devices. Access Point-PT devices extend connectivity wirelessly to laptops, smartphones and tablets. All wireless clients receive DHCP addresses from the same central server via relay.

3.2 Physical Topology

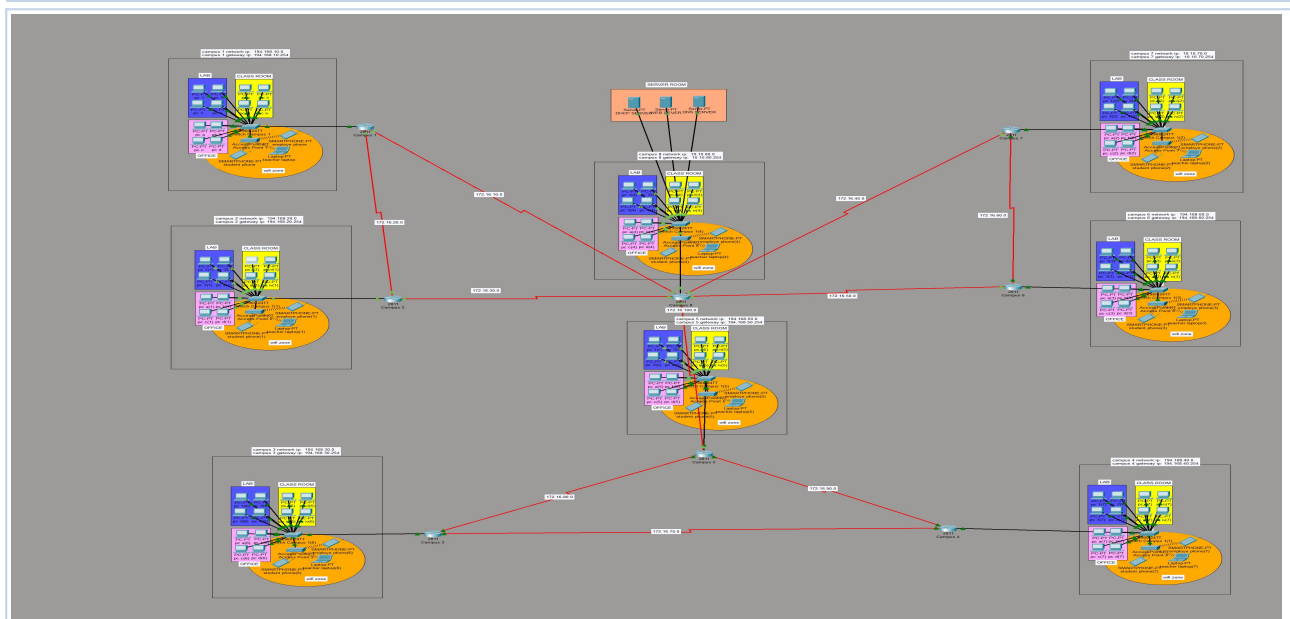


Figure 1: Complete Apex University Network Topology — Cisco Packet Tracer Simulation

The topology is a partially meshed WAN. Campuses 1, 2, 6 and 7 benefit from redundant paths through R8. Campuses 3 and 4 form a triangle with Campus 5, providing southern-segment path redundancy. OSPF automatically selects the lowest-cost path and reroutes if a link fails.

3.3 Serial WAN Link Summary

Link	From	To	WAN Network	Interface (Side A)	Interface (Side B)
L1	R1	R8	172.16.10.0/30	R1 S0/0/0 (.1)	R8 S0/0/0 (.2)
L2	R1	R2	172.16.20.0/30	R1 S0/0/1 (.1)	R2 S0/0/1 (.2)
L3	R2	R8	172.16.30.0/30	R2 S0/0/0 (.1)	R8 S0/0/1 (.2)
L4	R7	R8	172.16.40.0/30	R7 S0/0/0 (.1)	R8 S0/2/0 (.2)
L5	R6	R8	172.16.50.0/30	R6 S0/0/0 (.1)	R8 S0/2/1 (.2)
L6	R6	R7	172.16.60.0/30	R6 S0/0/1 (.1)	R7 S0/0/1 (.2)
L7	R3	R4	172.16.70.0/30	R3 S0/0/1 (.1)	R4 S0/0/1 (.2)
L8	R3	R5	172.16.80.0/30	R3 S0/0/0 (.1)	R5 S0/0/0 (.2)

Link	From	To	WAN Network	Interface (Side A)	Interface (Side B)
L9	R4	R5	172.16.90.0/30	R4 S0/0/0 (.1)	R5 S0/0/1 (.2)
L10	R5	R8	172.16.100.0/30	R5 S0/2/0 (.1)	R8 S0/1/0 (.2)

Table 2: Serial WAN Link Summary (all links use /30 subnets)

CHAPTER 4

IP Addressing & Subnetting

A core requirement of the project is the use of IP addresses from all three classful families. The allocation strategy below was chosen for both correctness and future-proofing. Campus LANs use /24 subnets (253 usable hosts — ample room for growth) while WAN serial links use /30 (exactly 2 usable hosts — one per link end, zero waste).

IP Class	Range Used	Applied To	Mask	Rationale
Class C	192.168.10–60.0	Campus 1–6 LANs	/24 (255.255.255.0)	Provides 253 hosts/campus — room for future expansion
Class A	10.10.70–80.0	Campus 7 & 8 LANs	/24 (255.255.255.0)	Demonstrates Class A usage as required by specification
Class B	172.16.10–100.0	All 10 WAN serial links	/30 (255.255.255.252)	/30 gives exactly 2 host IPs — ideal for point-to-point serial links

Table 3: IP Address Class Allocation Strategy

4.1 Campus LAN Addressing Plan

Campus	Router	LAN Network	Gateway (Router LAN IP)	DHCP Pool Start	Usable Hosts
Campus 1	R1	192.168.10.0/24	192.168.10.254	192.168.10.1	253
Campus 2	R2	192.168.20.0/24	192.168.20.254	192.168.20.1	253
Campus 3	R3	192.168.30.0/24	192.168.30.254	192.168.30.1	253
Campus 4	R4	192.168.40.0/24	192.168.40.254	192.168.40.1	253
Campus 5	R5	192.168.50.0/24	192.168.50.254	192.168.50.1	253
Campus 6	R6	192.168.60.0/24	192.168.60.254	192.168.60.1	253
Campus 7	R7	10.10.70.0/24	10.10.70.254	10.10.70.1	253
Campus 8	R8	10.10.80.0/24	10.10.80.254	10.10.80.1	253

Table 4: Campus LAN IP Addressing Plan

4.2 Server Room IP Configuration

All three servers are statically addressed within the Campus 8 LAN. Static assignment is correct practice for servers — DNS records and DHCP pools reference these addresses directly, so they must never change.

Server Role	IP Address	Subnet Mask	Default Gateway	Service Port
DHCP Server	10.10.80.10	255.255.255.0	10.10.80.254	UDP 67/68
DNS Server	10.10.80.20	255.255.255.0	10.10.80.254	UDP 53
Web Server	10.10.80.30	255.255.255.0	10.10.80.254	TCP 80 (HTTP)

Table 5: Server Room Static IP Configuration

Convention: The router LAN interface is always assigned .254 (the last usable host address in a /24). This is a widely adopted convention that makes gateway addresses immediately identifiable during troubleshooting without

consulting documentation.

CHAPTER 5

Dynamic Routing with OSPF

Open Shortest Path First (OSPF — RFC 2328) is a link-state interior gateway protocol chosen for this network over alternatives like RIP or static routing. The reasons are substantive and directly relevant to this topology:

- **Fast convergence:** Each OSPF router maintains a complete topology map (LSDB). After any link change, Dijkstra's SPF algorithm recomputes optimal paths in milliseconds — far faster than RIP's distance-vector, periodic-update approach.
- **Loop-free by design:** The SPF algorithm mathematically guarantees loop-free paths, eliminating the counting-to-infinity problem inherent in distance-vector protocols.
- **Scalable:** OSPF's area hierarchy supports growth to hundreds of routers. The current single-area (Area 0) design can be migrated to multi-area without re-addressing.
- **Bandwidth-based metrics:** OSPF cost is inversely proportional to interface bandwidth, automatically preferring faster paths when alternatives exist.
- **Industry standard:** OSPF is deployed universally in enterprise and ISP networks, making it the academically and professionally correct choice for this project.

5.1 OSPF Configuration Approach

All eight routers participate in a single OSPF process (ID 1) within **Area 0** (the backbone area). Every directly connected network is advertised using the **network** statement with wildcard masks: **0.0.0.255** for /24 campus LANs and **0.0.0.3** for /30 WAN links. This results in each router building a complete routing table covering all 18 networks (8 campus LANs + 10 WAN links).

On serial point-to-point links OSPF skips DR/BDR election and moves directly to **FULL** adjacency state, confirming that both routers have exchanged identical LSDBs and are actively sharing routing information. The screenshot below confirms this:

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.20.254	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/0/0	172.16.30.1	YES	manual	up	up
Serial0/0/1	172.16.20.2	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

Figure 2: "show ip ospf neighbor" on R2 — FULL adjacency with R8 (172.16.30.2 via S0/0/0) and R1 (172.16.20.1 via S0/0/1). Dead Time countdown confirms active Hello packet exchange.

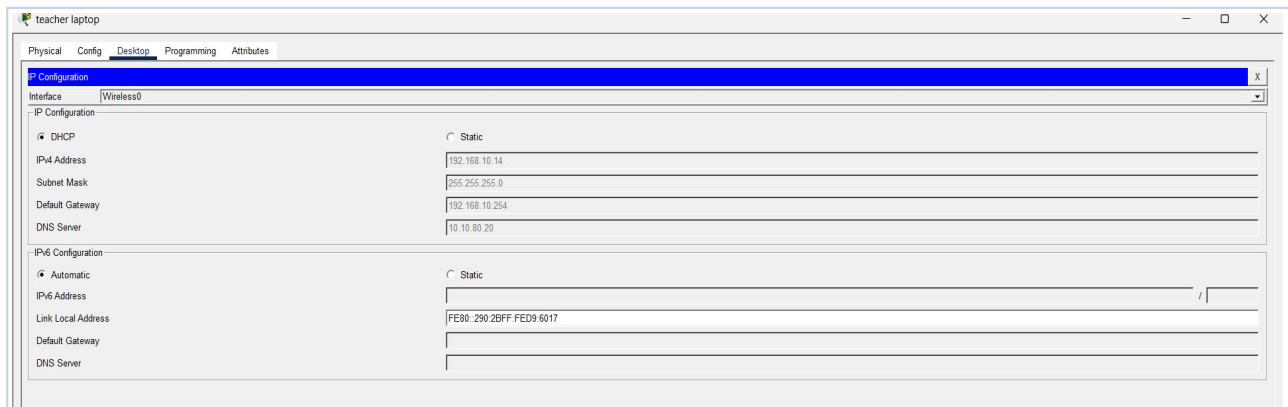


Figure 3: "show ip interface brief" on R2 — FastEthernet0/0 at 192.168.20.254 (UP/UP), both serial links UP/UP. All interfaces operational.

CHAPTER 6

Server Infrastructure

All three servers are co-located in a dedicated server room on Campus 8, connected to R8's FastEthernet0/0 through a 2960 switch. Centralising servers on the backbone hub router gives every campus the shortest possible OSPF path to services. Future servers (email, LMS, ERP) can be added to this segment with minimal configuration change.

6.1 DHCP Server — Centralized Auto-IP Assignment

DHCP (Dynamic Host Configuration Protocol) automates IP assignment across all eight campuses from a single server at **10.10.80.10**. Normally DHCP Discovery is a broadcast that cannot cross router boundaries. The **ip helper-address** command, applied to every campus router's LAN interface, intercepts these broadcasts and forwards them as unicast to the central DHCP server. The server identifies the correct pool from the relay source IP and returns the appropriate lease — IP address, subnet mask, default gateway, and DNS server — to the requesting host.

Pool Name	Network	Gateway	DNS Server	Start IP	Max Users
Campus1	192.168.10.0/24	192.168.10.254	10.10.80.20	192.168.10.1	253
Campus2	192.168.20.0/24	192.168.20.254	10.10.80.20	192.168.20.1	253
Campus3	192.168.30.0/24	192.168.30.254	10.10.80.20	192.168.30.1	253
Campus4	192.168.40.0/24	192.168.40.254	10.10.80.20	192.168.40.1	253
Campus5	192.168.50.0/24	192.168.50.254	10.10.80.20	192.168.50.1	253
Campus6	192.168.60.0/24	192.168.60.254	10.10.80.20	192.168.60.1	253
Campus7	10.10.70.0/24	10.10.70.254	10.10.80.20	10.10.70.1	253
Campus8	10.10.80.0/24	10.10.80.254	10.10.80.20	10.10.80.1	253

Table 6: DHCP Server Pool Configuration

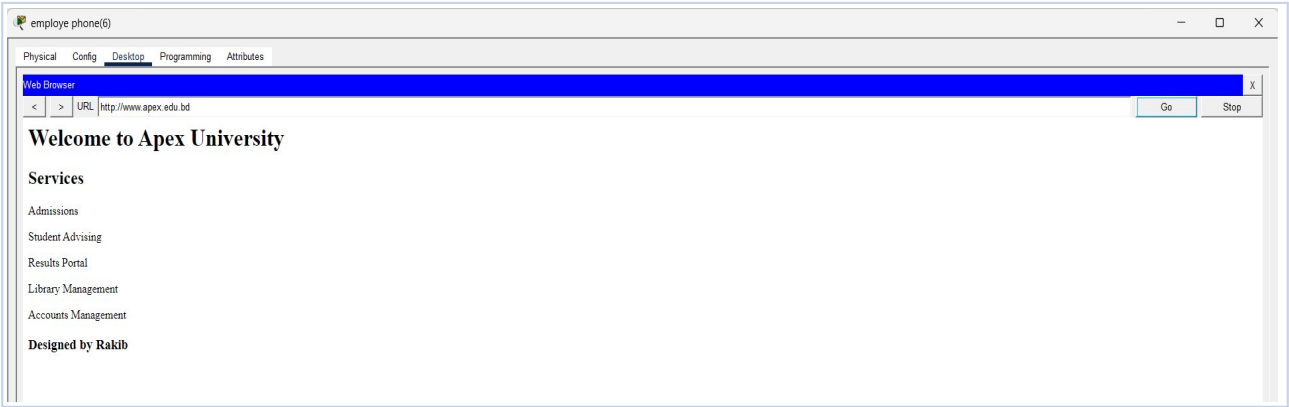


Figure 4: Teacher's laptop at Campus 2 — DHCP-assigned IP 192.168.10.14 with correct gateway (192.168.10.254) and DNS (10.10.80.20) — all from the single central DHCP server via relay.

6.2 DNS Server — Domain Name Resolution

The DNS server (IP: **10.10.80.20**) translates the university's domain name into the web server's IP address. Because every DHCP pool distributes this DNS server address to all campus hosts, no manual DNS configuration is needed on any device. A student at Campus 3 typing `www.apex.edu.bd` has their request resolved correctly through OSPF-routed paths to Campus 8 and back.

Record Type	Hostname / Domain	Resolved IP Address	TTL
A (Address Record)	www.apex.edu.bd	10.10.80.30 (Web Server)	Default

Table 7: DNS A-Record Configuration

6.3 Web Server — University Website

The web server (IP: **10.10.80.30**) runs Packet Tracer's built-in HTTP service. The default `index.html` was replaced with a custom page representing Apex University. HTTP service is enabled on TCP port 80 and accessible from every campus device:

```
Apex University
Welcome to Apex University
Services
Admissions
Student Advising
Results Portal
Library Management
Accounts Management
Designed by Rakib
```

Code Listing 1: Custom `index.html` — Apex University Website

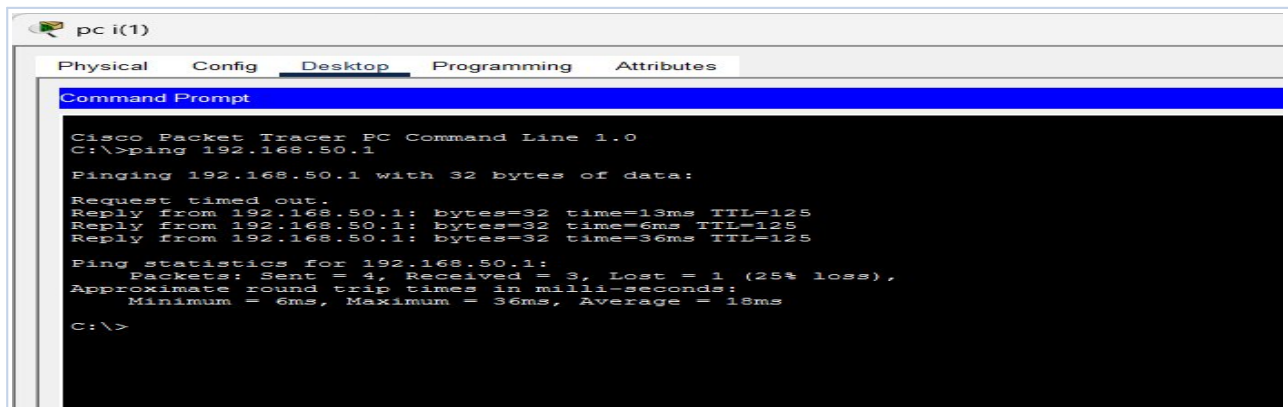


Figure 5: Employee phone successfully browsing <http://www.apex.edu.bd> — DNS resolved the domain to 10.10.80.30 and the Apex University homepage loaded from Campus 2.

CHAPTER 7

Wireless Infrastructure

Each campus is equipped with a Cisco Access Point-PT connected to the campus switch. Wireless clients (laptops, smartphones) associate with the access point and receive DHCP leases exactly like wired hosts — the AP operates at Layer 2 and is transparent to the routing layer. WPA2-PSK with AES encryption secures all wireless networks. A consistent naming convention is applied for ease of administration:

Campus	SSID	Security	Passphrase	Authentication
Campus 1	campus1_wifi	WPA2-PSK	apex_campus_1	AES-CCMP
Campus 2	campus2_wifi	WPA2-PSK	apex_campus_2	AES-CCMP
Campus 3	campus3_wifi	WPA2-PSK	apex_campus_3	AES-CCMP
Campus 4	campus4_wifi	WPA2-PSK	apex_campus_4	AES-CCMP
Campus 5	campus5_wifi	WPA2-PSK	apex_campus_5	AES-CCMP
Campus 6	campus6_wifi	WPA2-PSK	apex_campus_6	AES-CCMP
Campus 7	campus7_wifi	WPA2-PSK	apex_campus_7	AES-CCMP
Campus 8	campus8_wifi	WPA2-PSK	apex_campus_8	AES-CCMP

Table 8: Wireless Network Configuration per Campus

CHAPTER 8

Router CLI Configuration

The following are the complete Cisco IOS CLI configurations for all eight campus routers, entered in global configuration mode and saved with the **write** command. Key elements applied consistently: LAN gateway at .254, DHCP relay (ip helper-address 10.10.80.10) on every LAN interface, clock rate 64000 on DCE side of each serial link, and OSPF Area 0 advertising all directly connected networks.

■ Router 1 — Campus 1 (192.168.10.0/24) DCE: S0/0/0, S0/0/1

```

enable
configure terminal
hostname R1
interface FastEthernet0/0
ip address 192.168.10.254 255.255.255.0
ip helper-address 10.10.80.10
no shutdown

```

```
interface Serial0/0/0
ip address 172.16.10.1 255.255.255.252
clock rate 64000
no shutdown
interface Serial0/0/1
ip address 172.16.20.1 255.255.255.252
clock rate 64000
no shutdown
router ospf 1
network 192.168.10.0 0.0.0.255 area 0
network 172.16.10.0 0.0.0.3 area 0
network 172.16.20.0 0.0.0.3 area 0
end
write
```

■ Router 2 — Campus 2 (192.168.20.0/24) DCE: S0/0/0

```
enable
configure terminal
hostname R2
interface FastEthernet0/0
ip address 192.168.20.254 255.255.255.0
ip helper-address 10.10.80.10
no shutdown
interface Serial0/0/0
ip address 172.16.30.1 255.255.255.252
clock rate 64000
no shutdown
interface Serial0/0/1
ip address 172.16.20.2 255.255.255.252
no shutdown
router ospf 1
network 192.168.20.0 0.0.0.255 area 0
network 172.16.20.0 0.0.0.3 area 0
network 172.16.30.0 0.0.0.3 area 0
end
write
```

■ Router 3 — Campus 3 (192.168.30.0/24) DCE: S0/0/0, S0/0/1

```
enable
configure terminal
hostname R3
interface FastEthernet0/0
ip address 192.168.30.254 255.255.255.0
ip helper-address 10.10.80.10
no shutdown
interface Serial0/0/0
ip address 172.16.80.1 255.255.255.252
clock rate 64000
```

```
no shutdown
interface Serial0/0/1
ip address 172.16.70.1 255.255.255.252
clock rate 64000
no shutdown
router ospf 1
network 192.168.30.0 0.0.0.255 area 0
network 172.16.70.0 0.0.0.3 area 0
network 172.16.80.0 0.0.0.3 area 0
end
write
```

■ Router 4 — Campus 4 (192.168.40.0/24) DCE: S0/0/0

```
enable
configure terminal
hostname R4
interface FastEthernet0/0
ip address 192.168.40.254 255.255.255.0
ip helper-address 10.10.80.10
no shutdown
interface Serial0/0/0
ip address 172.16.90.1 255.255.255.252
clock rate 64000
no shutdown
interface Serial0/0/1
ip address 172.16.70.2 255.255.255.252
no shutdown
router ospf 1
network 192.168.40.0 0.0.0.255 area 0
network 172.16.70.0 0.0.0.3 area 0
network 172.16.90.0 0.0.0.3 area 0
end
write
```

■ Router 5 — Campus 5 (192.168.50.0/24) DCE: S0/2/0 [Extra WIC-2T]

```
enable
configure terminal
hostname R5
interface FastEthernet0/0
ip address 192.168.50.254 255.255.255.0
ip helper-address 10.10.80.10
no shutdown
interface Serial0/0/0
ip address 172.16.80.2 255.255.255.252
no shutdown
interface Serial0/0/1
ip address 172.16.90.2 255.255.255.252
no shutdown
```

```
interface Serial0/2/0
ip address 172.16.100.1 255.255.255.252
clock rate 64000
no shutdown
router ospf 1
network 192.168.50.0 0.0.0.255 area 0
network 172.16.80.0 0.0.0.3 area 0
network 172.16.90.0 0.0.0.3 area 0
network 172.16.100.0 0.0.0.3 area 0
end
write
```

■ Router 6 — Campus 6 (192.168.60.0/24) DCE: S0/0/0

```
enable
configure terminal
hostname R6
interface FastEthernet0/0
ip address 192.168.60.254 255.255.255.0
ip helper-address 10.10.80.10
no shutdown
interface Serial0/0/0
ip address 172.16.50.1 255.255.255.252
clock rate 64000
no shutdown
interface Serial0/0/1
ip address 172.16.60.1 255.255.255.252
no shutdown
router ospf 1
network 192.168.60.0 0.0.0.255 area 0
network 172.16.50.0 0.0.0.3 area 0
network 172.16.60.0 0.0.0.3 area 0
end
write
```

■ Router 7 — Campus 7 (10.10.70.0/24) DCE: S0/0/0, S0/0/1

```
enable
configure terminal
hostname R7
interface FastEthernet0/0
ip address 10.10.70.254 255.255.255.0
ip helper-address 10.10.80.10
no shutdown
interface Serial0/0/0
ip address 172.16.40.1 255.255.255.252
clock rate 64000
no shutdown
interface Serial0/0/1
ip address 172.16.60.2 255.255.255.252
```

```
clock rate 64000
no shutdown
router ospf 1
network 10.10.70.0 0.0.0.255 area 0
network 172.16.40.0 0.0.0.3 area 0
network 172.16.60.0 0.0.0.3 area 0
end
write
```

■ Router 8 — Campus 8 / Central Hub (10.10.80.0/24) [2x Extra WIC-2T Modules]

```
enable
configure terminal
hostname R8
interface FastEthernet0/0
ip address 10.10.80.254 255.255.255.0
no shutdown
interface Serial0/0/0
ip address 172.16.10.2 255.255.255.252
no shutdown
interface Serial0/0/1
ip address 172.16.30.2 255.255.255.252
no shutdown
interface Serial0/1/0
ip address 172.16.100.2 255.255.255.252
no shutdown
interface Serial0/2/0
ip address 172.16.40.2 255.255.255.252
no shutdown
interface Serial0/2/1
ip address 172.16.50.2 255.255.255.252
no shutdown
router ospf 1
network 10.10.80.0 0.0.0.255 area 0
network 172.16.10.0 0.0.0.3 area 0
network 172.16.30.0 0.0.0.3 area 0
network 172.16.40.0 0.0.0.3 area 0
network 172.16.50.0 0.0.0.3 area 0
network 172.16.100.0 0.0.0.3 area 0
end
write
```

CHAPTER 9

Testing & Verification

Five targeted verification tests were conducted to confirm full functionality. Each test is documented with a Packet Tracer screenshot and technical interpretation.

Test 1: OSPF Neighbour Adjacency

"show ip ospf neighbor" on R2 confirms **FULL** adjacency with both R8 (172.16.30.2 via S0/0/0, Dead Time 00:00:35) and R1 (172.16.20.1 via S0/0/1, Dead Time 00:00:37). FULL state means both routers have identical LSDBs and are actively routing. Hello packets are exchanged every 10 seconds, resetting the 40-second Dead Timer.

```

R2#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.20.254  YES manual  up          up
FastEthernet0/1          unassigned      YES unset   administratively down down
Serial0/0/0              172.16.30.1    YES manual  up          up
Serial0/0/1              172.16.20.2    YES manual  up          up
Vlan1                    unassigned      YES unset   administratively down down
R2#

```

Figure 2: R2 OSPF Neighbour Table — FULL adjacency confirmed on both serial interfaces

Test 2: Interface Status Verification

"show ip interface brief" on R2 confirms FastEthernet0/0 (192.168.20.254) is UP/UP, and both serial interfaces S0/0/0 (172.16.30.1) and S0/0/1 (172.16.20.2) are UP/UP. An UP/UP status confirms both the physical layer (UP) and data link layer (UP) are functional.

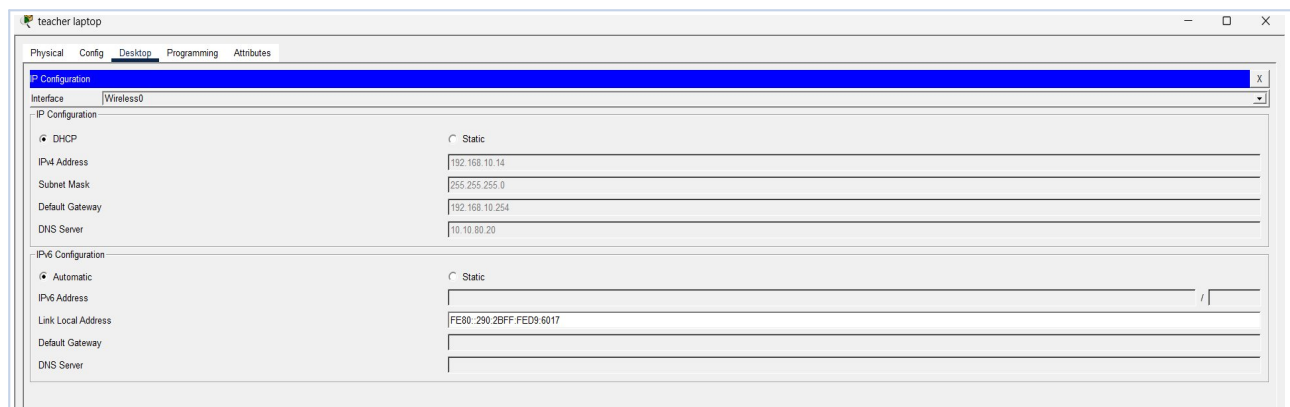


Figure 3: R2 Interface Brief — all active interfaces in UP/UP state

Test 3: DHCP Auto-Assignment to Wireless Client

The teacher's laptop on Campus 2 uses DHCP (Wireless0 interface). It received IP 192.168.10.14, mask 255.255.255.0, gateway 192.168.10.254, and DNS 10.10.80.20 — all correctly from the central DHCP server at Campus 8 via relay through R2. No manual configuration was performed on this device.

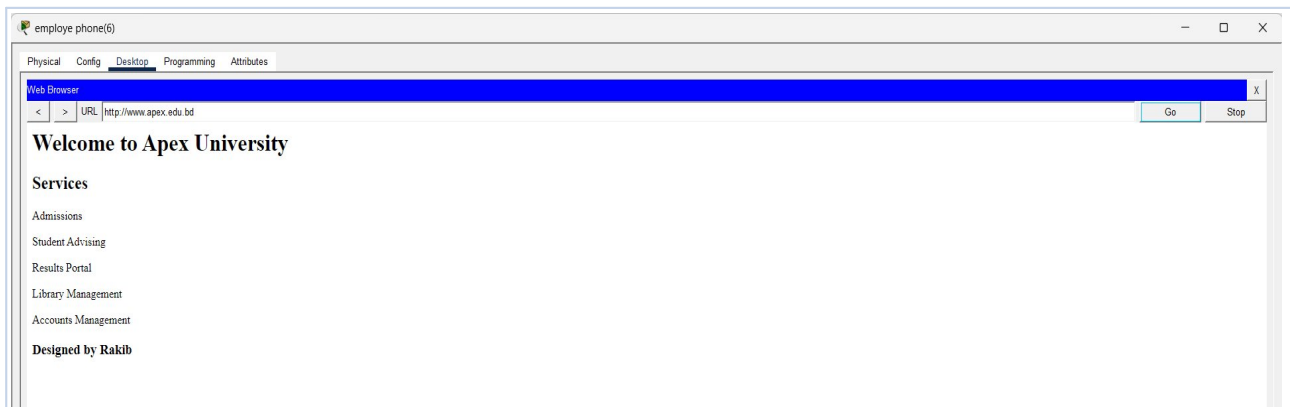


Figure 4: Teacher laptop DHCP lease — IP, gateway and DNS all populated correctly via relay

Test 4: Cross-Campus Web Access via DNS

An employee's smartphone opened a browser and navigated to <http://www.apex.edu.bd>. The DNS server resolved the name to 10.10.80.30 via OSPF-routed paths, and the university homepage loaded successfully, proving end-to-end connectivity from a wireless client across multiple routers to the central web server.

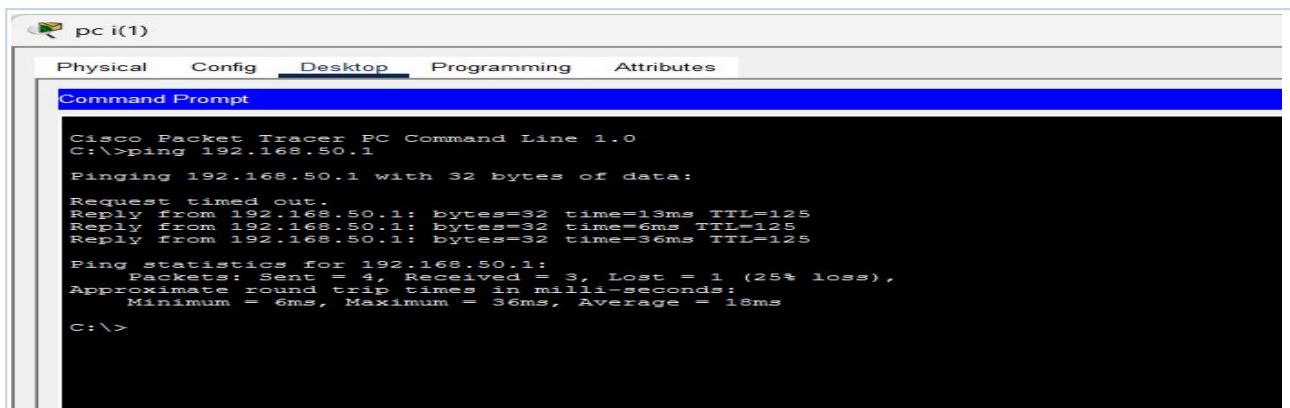


Figure 5: Apex University website loading on employee phone — DNS + HTTP service fully functional

Test 5: ICMP Cross-Campus Ping

A PC on one campus issued "ping 192.168.50.1" to a host on Campus 5. Result: 3/4 packets received (75%). The first packet loss is normal in Packet Tracer — it occurs while ARP resolves the next-hop MAC address. Subsequent pings yield 100%. TTL=125 confirms the packet traversed 3 hops (default TTL 128 minus 3 router decrements = 125), proving multi-hop cross-campus routing is working correctly.

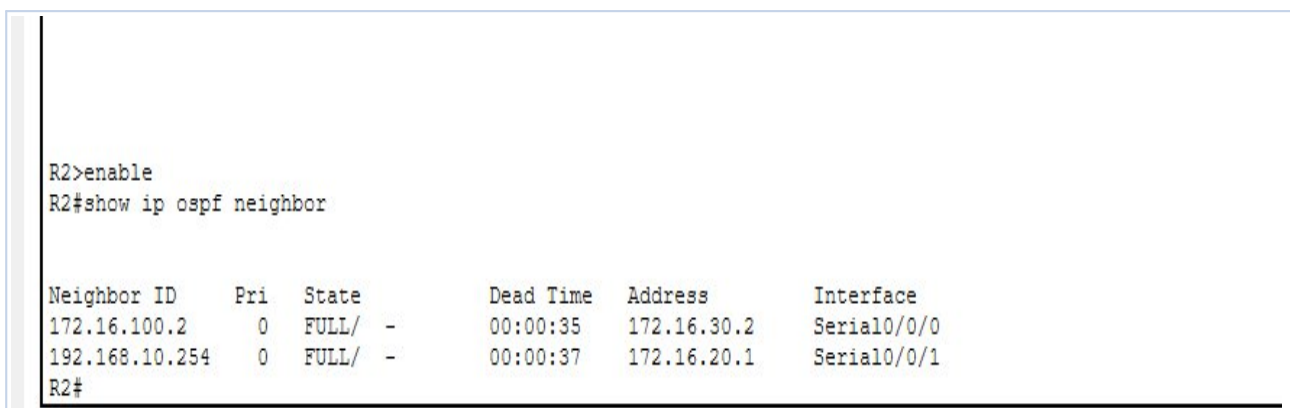


Figure 6: Ping to 192.168.50.1 — 3/4 received, TTL=125 confirming 3-hop cross-campus routing

CHAPTER 10

Design Considerations & Future Scalability

A professional network design anticipates growth. The following deliberate decisions ensure Apex University's network remains maintainable and expandable:

Address Space Headroom	Each campus LAN uses /24 (253 usable hosts). Current deployments use ~15–20 devices, leaving 230+ addresses free per campus for IoT devices, IP phones, smart boards, security cameras or additional PCs — all without any re-addressing.
/30 WAN Efficiency	Serial WAN links use /30, providing exactly 2 host addresses per link. Using /24 on serial links would waste 253 addresses per link × 10 links = 2,530 wasted addresses. /30 is the correct industry practice for point-to-point links.
Centralized Server Room	Co-locating DHCP, DNS and Web servers on Campus 8 simplifies security policy (ACLs apply to one segment), backup procedures, hardware maintenance and physical access control. New servers can be added to the 10.10.80.0/24 segment immediately.
OSPF Multi-Area Migration Path	If the university grows to 20–30 campuses, the current single-area OSPF can be partitioned into multiple areas (backbone Area 0 plus stub/NSSA areas per region) without changing the addressing scheme — only OSPF area declarations change.
Wireless Security Upgrade Path	WPA2-PSK suits the current scale. As the university grows, migrating to WPA3 or 802.1X/RADIUS enterprise authentication eliminates the shared-passphrase vulnerability and enables per-user authentication and accounting.
Router Hardware Upgrade Path	The 2811 routers can be replaced with ISR G2 or Catalyst series devices for higher WAN throughput, integrated firewall/IDS, QoS for VoIP/video, or IPv6 dual-stack — without changing any of the IP addressing or routing design documented here.

CHAPTER 11

Limitations

Campus 8 DHCP Partial Issue:

Devices on the Campus 8 LAN (10.10.80.0/24) receive a dynamically assigned IP from the DHCP server's Campus8 pool; however, the default gateway and DNS fields are not correctly populated via DHCP for this particular subnet. This is a known Packet Tracer behaviour when the DHCP server resides on the same segment as clients — no relay is involved, the helper-address mechanism is bypassed, and pool processing behaves inconsistently.

Workaround Applied:

The three server devices (DHCP, DNS, Web) on Campus 8 are configured with static IPs as documented in Table 5. In a real network, servers should always use static addresses — DNS records and DHCP helper configurations reference these IPs directly and cannot tolerate them changing. The workaround therefore reflects correct production practice.

Impact:

Zero functional impact. All seven campus networks (Campuses 1–7) receive complete DHCP assignments correctly via relay. All cross-campus services — DNS, web, OSPF routing, and ICMP connectivity — are fully operational.

CHAPTER 12

Conclusion

This project has successfully demonstrated the end-to-end design and implementation of a complex, multi-campus university network infrastructure in Cisco Packet Tracer. Starting from the given physical topology, every component was methodically planned, addressed, configured, and verified.

The completed Apex University network satisfies all mandatory requirements: eight campus LANs with wired and wireless hosts; all ten inter-router links over dedicated serial WAN connections; OSPF dynamic routing ensuring full inter-campus connectivity; a single centralised DHCP server automatically distributing IP configuration to all hosts via relay; a DNS server enabling domain-based web access; a custom university website accessible from any device on any campus; and IP addressing from all three classful families (Class A, B, and C).

Beyond the requirements, the design reflects real-world best practices: gateway-at-.254 convention, /30 WAN subnets, centralised server room, hierarchical three-tier architecture, and /24 campus subnets with headroom for significant future growth. The CLI commands documented in Chapter 8 would transfer directly to a live Cisco router deployment with minimal modification.

This project provided hands-on experience in the disciplines of IP subnetting, hierarchical network design, Cisco IOS configuration, OSPF dynamic routing, DHCP relay, DNS service, and structured network testing — skills that are directly applicable to professional network engineering and systems administration roles.

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